

MES Current-State Assessment & Target-State Blueprint

Assessment and future-state design for fictional regulated life sciences manufacturing

Current State

Target State

MES

Operating Model

Executive summary

Maps observed manufacturing reality to a cleaner MES-enabled target state across batch execution, QA review, equipment/material verification, deviations, and manufacturing visibility.

Problem statement

Manufacturing teams often inherit fragmented execution patterns: paper workarounds, unclear handoffs, disconnected review queues, and limited visibility into batch readiness.

Target-state recommendation

Define a target-state operating model that clarifies execution ownership, verification checkpoints, exception routing, review readiness, and system-of-record boundaries across MES, automation, quality, and data layers.

Observed manufacturing reality

- Operators need clearer prompts, references, and exception paths.
- QA review is slowed by incomplete context and manual reconciliation.
- QC, scheduling, and MES engineering often work from different readiness signals.

Target-state model

- ERP releases production demand and material context.
- MES orchestrates recipe execution, checks, signatures, and batch record state.
- QA/QC review uses structured exceptions, sample status, and release readiness indicators.

Recommendations

- Clarify ownership across operator, QA, QC, scheduling, and MES engineering roles.
- Standardize verification and exception handling.
- Create a shared MES-centered status model.

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Risks, dependencies, and KPI impact model

Risk and dependency framing

- Master data or recipe ownership gaps can slow adoption.
- Exception handling must be designed with QA and operations alignment.
- System integrations need clear data ownership and validation expectations.

Directional KPI / success measures		
Measure	Baseline / current signal	Target-state signal
Batch review readiness	Manual reconciliation required	Review package completeness visible earlier
Execution exceptions	Inconsistent routing and documentation	Standardized exception capture and triage
Operational visibility	Readiness status scattered across teams	Shared MES-centered readiness model

Values and examples are illustrative and directional. Validate assumptions with site-specific data before decision-making.